



A. LINEAR PROGRAMMING (LP)

Key Idea (Short Note)

Linear Programming helps us **maximize or minimize profit or cost** under given conditions.

Key Components

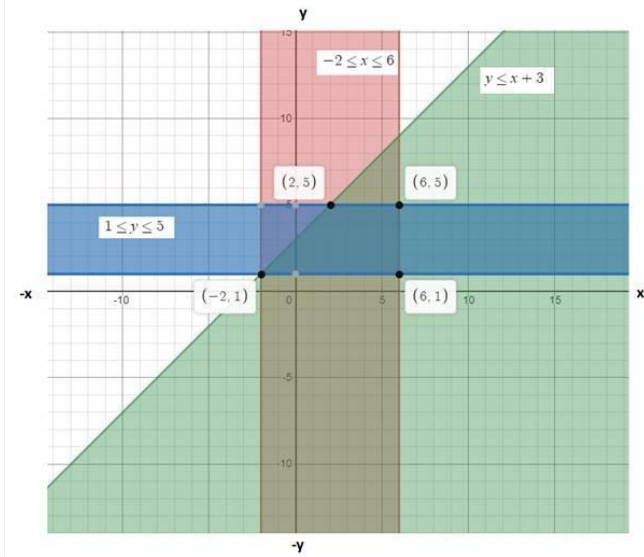
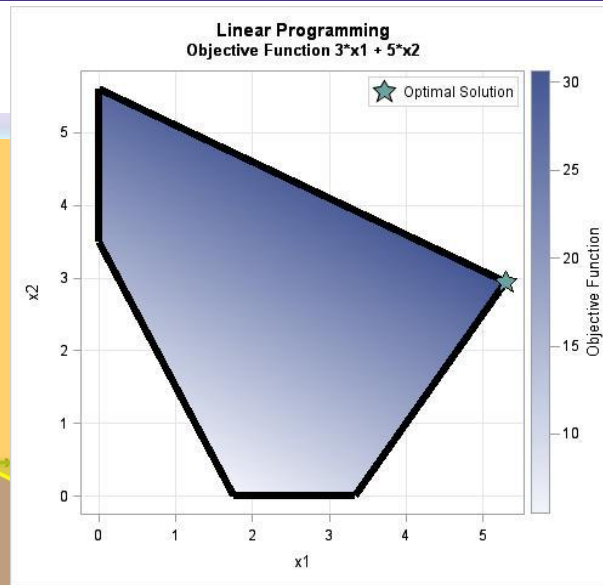
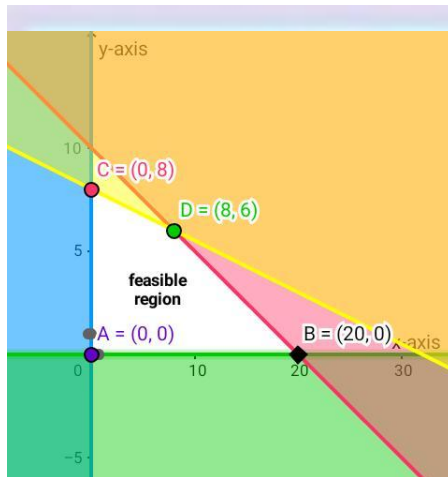
- Decision variables (x, y)

-

Objective function:

- $Z = ax + by$
- Constraints (inequalities)
- Non-negativity: $x \geq 0, y \geq 0$

Graphical Method



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Steps

1. Form equations
2. Plot lines
3. Shade feasible region
4. Find corner points
5. Test objective function

B. BEARINGS



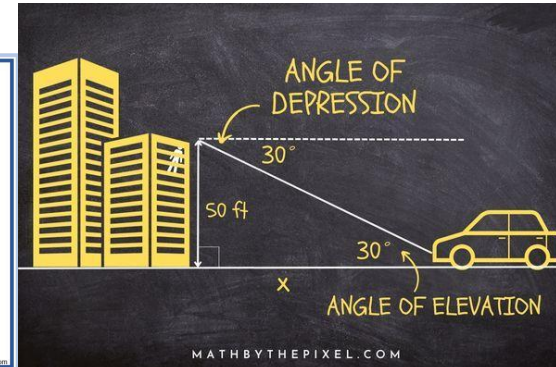
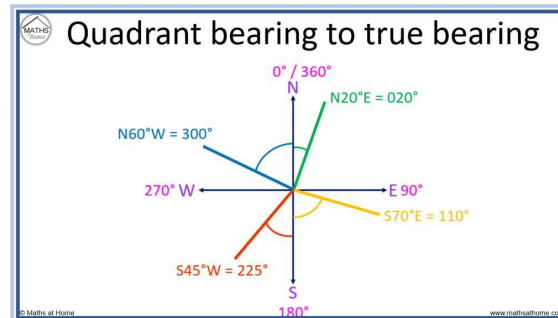
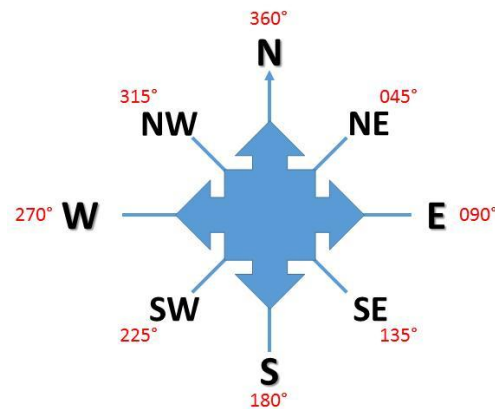
Short Note

Bearings show direction **measured clockwise from North (000° – 360°)**.

Rules

- Always measure from North
- Clockwise direction
- Back bearing:
 - Add 180° if $< 180^{\circ}$
 - Subtract 180° if $\geq 180^{\circ}$

Bearing Diagram



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C. ANGLES OF ELEVATION & DEPRESSION

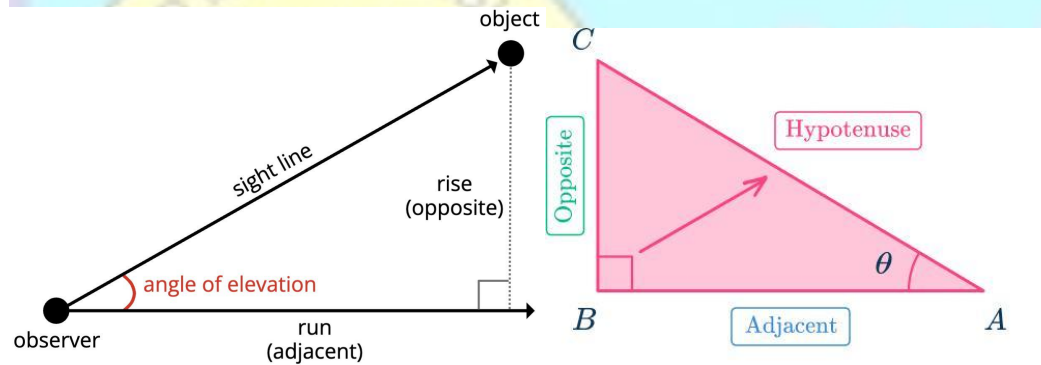
Short Note

- Elevation → looking up
- Depression → looking down
- Uses **right-angled triangles**

Key Formulas (SOHCAHTOA)

$$\sin \theta = \frac{\text{Opp}}{\text{Hyp}}, \cos \theta = \frac{\text{Adj}}{\text{Hyp}}, \tan \theta = \frac{\text{Opp}}{\text{Adj}} \quad \sin \theta = \frac{\text{Opp}}{\text{Hyp}}, \cos \theta = \frac{\text{Adj}}{\text{Hyp}}, \tan \theta = \frac{\text{Opp}}{\text{Adj}}$$

Visual Diagram



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D. FREQUENCY DISTRIBUTION

Short Note

Organizing data into tables for easier analysis.

Key Terms

- Class interval
- Class width
- Midpoint
- Frequency

Table Example

Frequency distribution and Histogram of Grouped Data

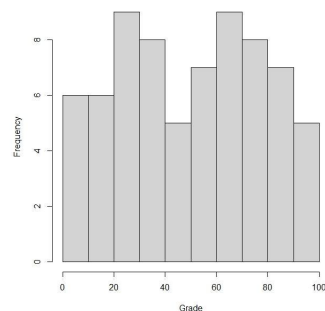
Problem :
In a Statistics test, the marks obtained by 70 students in a class is given as follows :

8 24 89 46 80 42 61 82 84 8 17 40 52 68 81
65 84 87 80 5 32 26 52 70 100 50 88 15 40 74
11 68 27 73 0 85 80 31 30 39 98 83 58 6 63
54 95 19 1 28 51 34 50 73 69 92 75 36 47 27
21 14 87 91 23 31 22 14 57 53

Solution :
Prepare a frequency distribution of the above data.

Frequency distribution :

Class Interval	frequency
0 - 10	6
10 - 20	6
20 - 30	9
30 - 40	8
40 - 50	5
50 - 60	7
60 - 70	9
70 - 80	8
80 - 90	7
90 - 100	5



Complete Frequency Table

Table 2-6
Grouped Frequency Distribution for the Test
Scores of 52 Students in Statistics

Class Intervals (ci)	Frequency (f)	Class Mark (x)	Class Boundary (cb)	Relative Frequency (rf)	<cf	>cf
0 - 2	20	1	-0.5 - 2.5	38.5%	20	52
3 - 5	14	4	2.5 - 5.5	26.9%	34	32
6 - 8	15	7	5.5 - 8.5	28.8%	49	18
9 - 11	2	10	8.5 - 11.5	3.8%	51	3
12 - 14	1	13	11.5 - 14.5	1.9%	52	1

Class interval	Tally marks	Frequency
0-10		4
10-20		7
20-30		5
30-40		10
40-50		5
50-60		8
60-70		5
70-80		8
80-90		5
90-100		3

E. OGIVES (CUMULATIVE CURVES)

Short Note

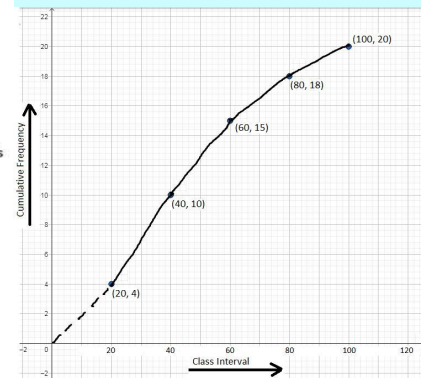
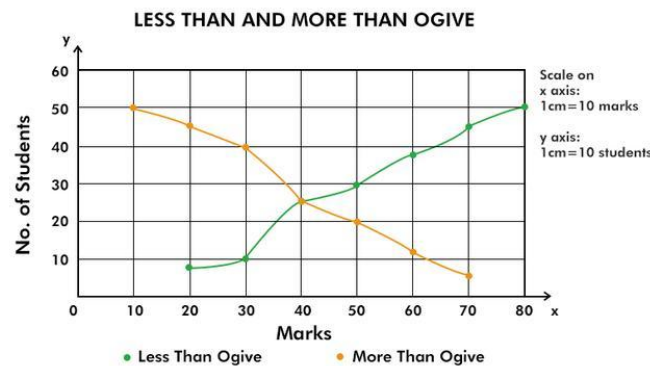
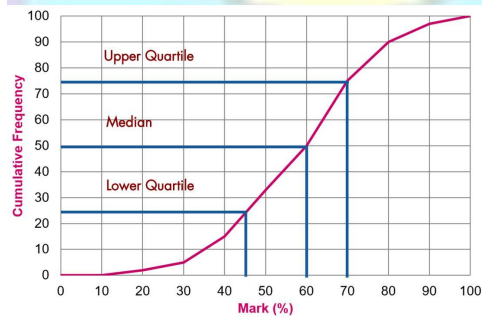


Graph used to show cumulative frequency.

Key Points

- X-axis → class boundaries
- Y-axis → cumulative frequency
- Used to find:
 - Median
 - Quartiles
 - Percentiles

Ogive Graph



F. MEASURES FROM FREQUENCY

Short Notes

Mean

$$\text{Mean} = \frac{\sum (f \times x)}{\sum f} \quad \text{Mean} = \frac{\sum (f \times x)}{\sum f}$$

Median

$$\text{Median} = L + \left(\frac{n/2 - CF_b}{f_m} \right) \times c \quad \text{Median} = L + \left(\frac{n/2 - CF_b}{f_m} \right) \times c$$

Mode

- Modal class = highest frequency

Comparison

- Mean → best for balanced data
- Median → best for extreme values
- Mode → most common value



G. PROBABILITY

Short Note

Probability measures likelihood of events.

Formula

$$P(E) = \frac{n(E)}{n(S)} \quad P(E) = \frac{n(E)}{n(S)}$$

Key Rules

- Complement: $P(A') = 1 - P(A)$ $P(A') = 1 - P(A)$
-

Mutually exclusive:

- $P(A \cup B) = P(A) + P(B)$ $P(A \cup B) = P(A) + P(B)$
-

Independent:

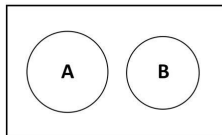
- $P(A \cap B) = P(A) \times P(B)$ $P(A \cap B) = P(A) \times P(B)$

Probability Illustration

Mutually Exclusive Events

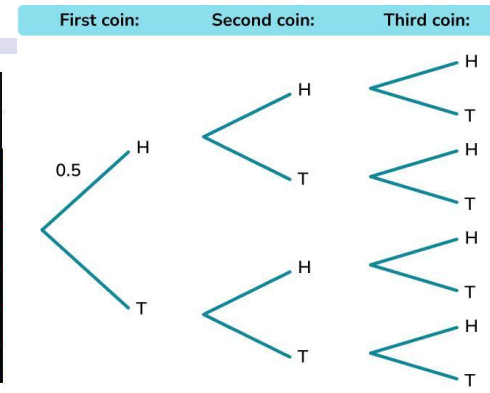
$$P(A \text{ or } B) = P(A) + P(B)$$

$$P(A \text{ and } B) = 0$$



Note: No overlap between A and B

		White Die					
		1	2	3	4	5	6
Red Die	1	(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1)
	2	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)
	3	(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3)
	4	(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4)
	5	(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)
	6	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)



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COMPREHENSIVE WORKED EXAMPLES (SIMPLIFIED)

Example 1 (Linear Programming)

- Maximize: $P = 5000x + 3000y$
- Solution → Best at (0,100)
- Profit = ₦300,000

Example 2 (Bearings + Trig)



Final bearing $\approx 077^\circ$

Example 3 (Statistics)

- Find mean, median
- Draw ogive

Example 4 (Probability)

$$P = \frac{11}{36} \quad P = \frac{11}{36}$$

COMPREHENSIVE EVALUATION QUESTIONS

1. Formulate LP problem
2. Convert bearings
3. Solve elevation problem
4. Calculate mean
5. Draw ogive
6. Solve probability
7. Use probability rules
8. Find back bearing
9. Find quartiles
10. Choose best measure



ASSIGNMENT

1. Solve LP problem (cakes & bread)
2. Navigation problem
3. Trigonometry heights
4. Statistics + ogive + quartiles
5. Probability with cards

